Project Scope

Our project is a smart irrigation controller that decides when to water a garden based on soil moisture and simulated rainfall. Our controller will check the soil moisture and a rain forecast before deciding whether to water. We decided to build it as a C++ program and compare it to a basic watering timer to show how much water it saves. This project will use a test-based interface and simulated data, so no physical hardware is needed.

Expected Deliverables

- Iteration 1: Project Plan
- Soil moisture and rain simulation
- Smart controller and timer controller
- Text-based results and comparison between the two controllers
- GitHub repository with code and README
- Final report
- In-class presentation with demo

Team Objectives

1. Building a working C++ program that simulates soil moisture and rainfall
2. Create a smart controller that only waters when the soil is too dry and rain is not expected
3. Show that the smart controller uses less water than the timer controller
4. Provide a text-based menu so users can run either controller and compare results
5. Test all features and document the project in a README

Individual Objectives

Claire - Simulation

1. Build the soil moisture simulation and the rain simulation
2. Save results to a CSV file and calculate total water used
3. Contribution: creates and provides the simulated garden that the controllers make decisions about

Nyesha - Controllers and Interface

1. Build timer controller and smart controller
2. Build text-based menu and results display
3. Contribution: builds the decision making part and how users interact with it

Both

1. Test each other's code and review changes on GitHub
2. Write README, final report, and presentation

Technologies and Tools

- C++ → The main language for the project.
- g++ compiler → This will compile our program.
- VS Code → This is our code editor.
- Git/GitHub → This is how we share code and look at all our versions.
- Excel → This will open the CSV output and make our charts.
- Google Docs → This is where we track our tasks and deadlines.
- PowerPoint → This is where we make our final presentation.

Project Timeline

Iteration	Date	Task	Who?	Deliverable
1. Planning	10/13/2026	Define objectives, choose our tools, write project plan	Both	Project plan on GitHub
2. Simulation	10/23/2026	Soil simulation, rain simulation/forecast	Claire	Program simulates 30+ days of moisture/rain
		Timer controller, basic text menu	Nyesha	Timer controller runs from menu
3. Smart Controller	11/06/2026	Results display, CSV output	Claire	Results and CSV file
		Smart controller, settings menu	Nyesha	Smart controller runs
4. Testing	11/20/2026	Testing all features, debugging, making charts	Both	Working final program and results
5. Final	12/01/2026	README, final report, presentation	Both	Final report, completed repo, and presentation

Milestones

- 10/13/2026 → Iteration 1 submitted
- 10/23/2026 → Simulation working
- 10/06/2026 → Both controllers working

- 11/20/2026 → Program fully tested and debugged
- 12/01/2026 → Final report and presentation submitted

Communication/Progress Tracking/Challenges

We will plan to meet in person or online once a week to review our progress and plan for each iteration. We will track our tasks and deadlines on google docs, documenting our progress. We will check each other's code before it is added to GitHub. If either of us are stuck on our part, we will let the other know and try to work on it together. If we need more time for a deadline, we will communicate and let the other know. If we disagree on the code or decisions, we will discuss with each other and try to come to a conclusion.

Contribution Table

Who?	**Completion**	**Responsibility**
Claire	Completed project plan/iteration 1	Work on soil simulation and rain simulation
Nyesha	Completed project plan/iteration 1	Work on timer controller and basic menu

Meeting 1: October 13, 2026

- Decided to build the program in C++ with a text-based interface and simulated data.
- Agreed on team and individual objectives.
- Created the project timeline and Iteration 1